

Async FIFO (Gray)

A synchronous FIFO shares one clock

Empty starter

- Starter: depth-eight FIFO, empty equals one, full equals zero
- Write domain has wbin and wgray; read domain has rbin and rgray
- Step wclk to write wdata hex A5 if not full, wbin advances, wgray updates
- Remote rgray syncs through rq1 and rq2 into the write domain for full detection
- Step rclk on the read side, wq1 and wq2 sync wgray for empty
- After writes and sync settle, empty clears and a read returns the oldest byte

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Async FIFO (Gray pointers)

Cross-clock FIFO sketch: binary counters for addressing, **Gray** pointers through 2-FF sync for empty/full.

Starter example: empty depth-8 FIFO — write on wclk, read on rclk; watch Gray sync and empty/full.

Load starter example

Challenges (1/22)

Quiz: **async**. FIFO with different read/write clocks is? Answer: **async**

Answer

Show hint

Check

Next

Idle

quiz-async

quiz-gray

quiz-2ff

quiz-empty

starter

wclk

rclk

gray-fn

gray-btn

not-empty

full

drain

sync2

demo

explain

code-gray

wgray-track

rq-pipe

reset

occ

rdata

full-path

Core ideas

Two clocks

Write and read domains free-run — no shared edge.

Gray sync

Only one bit changes per count — safer CDC of pointers.

Empty / full

Compare local Gray to 2-FF-synced remote Gray.

Async FIFO starter

learn_digital — async fifo

Workbook practice

- On paper, compute Gray for binary zero through three
- Draw write and read domains with two-FF synchronizers crossing between them
- Explain why binary pointers are unsafe for CDC
- Write the empty condition in words
- Trace one write on wclk and when the read side can safely see not-empty

Pitfalls to watch

- Do not compare unsynchronized binary pointers across clocks
- Empty and full can lag a few cycles after the other domain moves, that is intentional
- Gray is required because multi-bit binary can glitch during CDC
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, write once, sync, read once, then fill and drain
- On paper, list Gray codes for four binary values
- When you are ready, take the short quiz, then continue to handshake valid-ready